Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: Study of MK-4830 as Monotherapy and in Combination With Pembrolizumab (MK-3475) in Participants With Advanced Solid Tumors (MK-4830-001) **Short Title/Acronym:** MK-4830-001 **Protocol Number:** MK-4830-001 **NCT Number:** NCT03564691 **Sponsor/Company:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** MK-4830 (first-in-class anti-ILT4 monoclonal antibody), Pembrolizumab (Keytruda, anti-PD-1), and various combination chemotherapies. **Phase of Development:** PHASE1 **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and preliminary efficacy of MK-4830 as monotherapy and in combination with pembrolizumab (and other agents in specific expansion arms) in participants with advanced solid tumors, and to determine the Maximum Tolerated Dose (MTD) or Maximum Administered Dose (MAD). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS) based on RECIST v1.1, and the detailed Pharmacokinetics (PK) profile of MK-4830.

2.2 Background & Rationale MK-4830 is a novel human IgG4 monoclonal antibody targeting the immunoglobulin-like transcript 4 (ILT4) receptor. The rationale for this study is to reprogram immunosuppressive macrophages, thereby alleviating the myeloid-suppressive components of the tumor microenvironment. Modulating ILT4 is hypothesized to abrogate a known PD-1 resistance mechanism, making the combination of MK-4830 and pembrolizumab a highly promising immunotherapy approach for heavily pre-treated patients with advanced solid tumors who previously developed resistance to anti-PD-1/PD-L1 agents.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Uncontrolled (Multi-arm dose escalation, dose expansion, and coformulation arms A-N) **Blinding:** Open-label **Randomization:** Non-randomized (assignment to dose cohorts and specific expansion arms)

3.2 Planned Enrollment & Duration Targeted Enrollment: 470 participants **Number of Sites:** Multi-center global trial **Estimated Study Start:** 2018 **Trial Status:** COMPLETED

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically- or cytologically-confirmed advanced/metastatic solid tumor (Neoplasms) by pathology report. 2. Has received, has been intolerant to, or has been ineligible for all treatment known to confer clinical benefit. (Note: Specific expansion arms have unique prior therapy requirements, such as progression on a platinum-containing regimen like carboplatin or cisplatin [trade name: Platinol]). 3. Has measurable disease by Response Evaluation Criteria In Solid Tumors version 1.1 (RECIST 1.1), RANO, or modified RECIST (mRECIST). 4. Submits an evaluable baseline tumor sample for biomarker and tumor microenvironment analysis.

4.2 Exclusion Criteria 1. Has had chemotherapy, definitive radiation, or biological cancer therapy within 4 weeks (2 weeks for palliative radiation) prior to the first dose of study therapy. 2. Has not recovered from all radiation-related toxicities to Grade 1 or less, requires corticosteroids, or had radiation pneumonitis. 3. Has a history of a second malignancy, unless potentially curative treatment has been completed with no evidence of recurrence for 2 years. 4. Active autoimmune disease, clinical evidence of ascites/peritoneal metastases (in specific arms), or active infections requiring therapy.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-4830 administered at escalating doses as a monotherapy or in combination. In combination arms, MK-4830 is administered intravenously following pembrolizumab. A coformulation of MK-4830A (800 mg MK-4830 + 200 mg pembrolizumab) is also evaluated. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** 21-day cycles. Treatment may be administered for a maximum of 35 cycles (up to approximately 2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced cancer. Certain arms permit targeted combination chemotherapies (e.g., carboplatin, pemetrexed, lenvatinib, paclitaxel, trastuzumab). **Prohibited:** Concurrent unauthorized anti-cancer therapy, systemic immunosuppressants, and live-attenuated vaccines.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Biopsy, RECIST Imaging
Baseline	Day 1 (Cycle 1)	Assignment to dose cohort, First Dose of MK-4830 +/- Pembrolizumab
Interim	Every 21 Days	

(Q3W) | Safety assessments, PK blood draws, Efficacy evaluation via continuous RECIST grading | | **End of Study** | Up to Cycle 35 | End of treatment (approx. 2 years) or until disease progression, unacceptable toxicity, or withdrawal; followed by survival follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Dose-Limiting Toxicities (DLTs):** Incidence of specific hematologic and nonhematologic toxicities to establish safety and identify the Maximum Tolerated Dose (MTD). 2. **Number of Participants Who Experienced an Adverse Event (AE):** Tracking of any unfavorable/unintended signs occurring during the study. 3. **Number of Participants Who Discontinued Study Treatment Due to an AE.**

7.2 Secondary & Exploratory Endpoints 1. **Area Under the Curve (AUC)** of Plasma MK-4830. 2. **Minimum Concentration Between Doses (Ctrough)** of Plasma MK-4830. 3. **Maximum Drug Concentration (Cmax)** of Plasma MK-4830. 4. Objective Response Rate (ORR), Duration of Response (DOR), and Progression-Free Survival (PFS) via RECIST v1.1 or RANO.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous AE tracking at every Q3W cycle visit. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** A Safety Review Committee utilizes a modified toxicity probability interval (mTPI) design framework to actively guide and adjudicate dose-escalation decisions.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All-Patients-as-Treated (APaT) / Safety Set for participants receiving at least one dose of study treatment. **Sample Size Justification:** Target enrollment of 470 participants across dose escalation and expansion arms, formulated to efficiently identify the safe Recommended Phase 2 Dose (RP2D) and test combinations. **Handling of Missing Data:** Time-to-event endpoints (DOR, PFS) evaluated using standard Kaplan-Meier estimates and appropriate censoring criteria.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Periodic onsite and remote source data verification by the sponsor. **Audit Trail:** Secure Electronic Data Capture (eCRF), continuous safety tracking, and strict RECIST independent reviews.

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11. Study Identifiers & Governance

Primary ID: MK-4830-001 **Registry Number:** NCT03564691 **Sponsor/Lead Organization:** Merck **Study Title:** Study of MK-4830 as Monotherapy and in Combination With Pembrolizumab (MK-3475) in Participants With Advanced Solid Tumors (MK-4830-001) **Official Regulatory Title:** A Phase 1 Open Label, Multi-Arm, Multicenter Study of MK-4830 as Monotherapy and in Combination With Pembrolizumab for Participants With Advanced Solid Tumors

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12. Clinical Framework (Oncology Specific)

Disease Areas / Primary Condition: Neoplasms (Advanced Solid Tumors) **Diagnosis Threshold:** Histologically or cytologically confirmed advanced/metastatic disease **Medical Methodology:** Immunotherapy / Macrophage Reprogramming (Anti-ILT4 and Anti-PD-1 blockade) **Optimization Target:** Dose-escalation to establish Maximum Tolerated Dose (MTD) and RP2D

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13. Study Procedures & Methodology

Management Pathway: Standard Advanced Oncology Clinical Pathway / RECIST v1.1 Tumor Assessments **Education Protocol (HCP):** Site Initiation Visits, safety monitoring guidelines, and Investigator's Brochure (IB) **Education Protocol (Patient):** Informed consent process, reporting of AEs, and cycle scheduling **Quality Audit Mechanism:** Independent radiology review (IRR) of responses and standard clinical site monitoring

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14. Observation & Follow-Up Schedule

Total Duration: Up to 35 cycles (approx. 2 years) of treatment + subsequent survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) aligned with the 21-day cycles **Remote Monitoring Frequency:** Continuous per sponsor’s trial monitoring plan **Checklist Review Interval:** Safety, PK, and AE review at the start of each 21-day cycle

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**

Clinical Operations Lead | [Name] | ___ | *[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* | _____ | [DD-MMM-YYYY] |